LESSON 8.6

Solve and Graph Inequalities

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve an inequality and graph its solution set on a number line.

WHY THIS MATTERS Detective Kim knows that the suspect withdrew more than $50 from an ATM. The amount plus a $3 fee totaled more than $53. She writes x + 3 > 53 and needs to solve for x, then graph the solution set to share with the bank. How can she find and display all possible withdrawal amounts?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Solve *Spanish: Resolver*	To find the number that makes it true.	x + 3 > 10 → subtract 3 → x > 7 — any number greater than 7 works
Graph *Spanish: Graficar*	To show answers on a number line with circles and shading.	x > 7: open circle at 7, shade right — shows all solutions at a glance
Solution *Spanish: Solución*	A number that makes the equation or inequality true.	x = 8 is a solution to x > 7 because 8 > 7 is true; x = 5 is not because 5 > 7 is false
Substitute *Spanish: Sustituir*	To put a number in place of a letter.	Substitute x = 5 into x + 3 > 10: 5 + 3 = 8, and 8 > 10 is false, so x = 5 is not a solution
Solution set *Spanish: Conjunto solución*	All the numbers that make the inequality true.	For x > 4, the solution set is every number greater than 4, such as 5, 6, 7, ...
Inverse operation *Spanish: Operación inversa*	Two math actions that undo each other, like × and ÷.	To solve x + 3 > 7, use the inverse of adding 3: subtract 3 to get x > 4.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Solve x + 3 > 53. The inverse of adding 3 is subtracting 3.

1. I subtract 3 from both sides: x + 3 − 3 > 53 − 3, so x > 50.
2. The symbol > does not include 50, so I use an open circle at 50.
3. 'Greater than' shades right. Check: x = 60 gives 60 + 3 = 63 > 53. True!

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Solve $x - 5 \leq 8$. What is the inverse of subtracting 5?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Solving an inequality is like solving an equation because I still use _____, but it is different because the answer is _____ instead of _____.

Word bank Solve | Graph | Solution | Substitute | Solution set | Inverse operation | inverse operations | many values | one value | graph | solution set | number line

WATCH OUT!

A common mistake

A common mistake in Solve and Graph Inequalities is shading the number line in the wrong direction after solving. For example, a student solves $x + 4 > 10$ correctly to get $x > 6$, but then shades LEFT toward smaller numbers instead of RIGHT toward greater numbers. The shading direction always matches the solved symbol, not the original problem: '>' and '≥' shade right (greater numbers), '<' and '≤' shade left (smaller numbers)

LESSON 8.6

Solve and Graph Inequalities

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Solve: x + 6 > 14

(A) x > 8
(B) x < 8
(C) x = 8
(D) x > 20

STEPS TO TRY
1. Solve it like an equation: x has 6 added to it. The > symbol just rides along.
2. Subtract 6 from both sides — the inequality symbol keeps pointing the same way when you subtract.
3. Compute 14 − 6 and keep the >. Then test: pick a number bigger than your result and check it makes x + 6 > 14 true.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Solve: $x - 9 \leq 3$

(A) $x \leq 6$

(B) $x \leq 12$

(C) $x \geq 12$

(D) $x = 12$

STEPS TO TRY

1. x has 9 subtracted from it. Undo that first; the ≤ symbol keeps its direction.
2. Add 9 to both sides. Adding the same number to both sides never flips the symbol.
3. Compute 3 + 9 and keep the ≤. Check by testing a value at or below your boundary in the original inequality.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: $x + 10 \geq 25$

(A) $x \geq 15$

(B) $x \leq 15$

(C) $x = 15$

(D) $x \geq 35$

STEPS TO TRY

1. What's being done to x? 10 is added. Plan the inverse operation before you look at the choices.
2. Subtract 10 from both sides; the ≥ stays exactly as it is.
3. Compute 25 − 10 and keep the ≥. Test your boundary value — it should make both sides of the original exactly equal.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Solve: $x - 4 < 11$

(A) $x < 7$

(B) $x < 15$

(C) $x > 15$

(D) $x = 15$

STEPS TO TRY

1. x has 4 taken away. Which operation puts that 4 back?
2. Add 4 to both sides — the $<$ direction does not change when you add.
3. Compute $11 + 4$ and keep the $<$. Check: a number just below your boundary should make $x - 4 < 11$ true.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How is solving an inequality similar to and different from solving an equation?

SENTENCE STARTER Solving an inequality is like solving an equation because I still use ___, but it is different because the answer is ___ instead of ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help